

Note 2.1 Rate of Change and Tangents to Curves

Average and Instantaneous Speed

- Average Speed:
average speed = distance covered / time elapsed
- Instantaneous Speed:
Time elapse approached zero (limit idea)

Average Rates of Change and Secant Lines

- **Definition:**
The average rate of change for $y = f(x)$ with respect to x over $[x_1, x_2]$ is

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{f(x_1 + h) - f(x_1)}{h}, h \neq 0$$

- **Secant lines:**
A line joining two points of a curve

Instantaneous Rates of change and Tangent Lines

- **Definition:**
If $\frac{f(x_1 + h) - f(x_1)}{h}$ approaches m_0 **from either side**, then we say the tangent p exists and the slope of the tangent line is m_0 .